

An Alternative Proof of an Elementary Property of Pythagorean Triples

R. Scott McIntire

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1 Introduction

When I was a freshman in college I was briefly interested in the idea of using the areas of geometric figures to produce interesting formulas. The example that inspired me was the way one may derive the Pythagorean formula by examining a dissected square composed of an inscribed square and triangles.

I tried to do something similar with a more complicated geometric figure and I came up with figure 1.

By inscribing a circle¹ in a *right* triangle, one can decompose the triangle into two pairs of right triangles and a square. The fact that the sub-triangles are right triangles follows using the fact that any radial segment of a circle is perpendicular to its associated tangent line.

From the figure, the lower right four sided sub-figure has equal sides, r . I claim that this sub-figure is a square and not a rhombus. This can be seen as three of its interior angles are 90° ; and, as the interior angles of a four-sided *convex* polygon must sum to 360° , the last interior angle must also be 90° . Consequently, the four-sided figure is a square.

Below we find a well known relation between the sides, a , b , and c of the original triangle

¹Existence and uniqueness of the inscribed circle in a triangle is a standard fact: the angle bisectors of the triangle are concurrent at the incenter, which is then equidistant from the three sides.

when the sides are integers in *reduced form*.² This is done by investigating the fact that the sum of the areas of these component triangles and square must be the same as the area of the original triangle.

2 A Basic Property of Primitive Pythagorean Triples

Writing the equivalence of the area of the figure and its parts we have:

$$\overbrace{r^2}^{\text{Square}} + \overbrace{2 \frac{r(a-r)}{2}}^{\text{Two lower triangles}} + \overbrace{2 \frac{r(b-r)}{2}}^{\text{Two upper triangles}} = \overbrace{\frac{ab}{2}}^{\text{Original triangle}} \quad (1)$$

Simplifying, this is:

$$r^2 + r(a-r) + r(b-r) = \frac{ab}{2} \quad (2)$$

One can solve the quadratic for r (the quadratic formula gives $r = (a+b \pm c)/2$; only the minus sign is geometrically valid, since $r < a$ and $r < b$) or read off from the figure that $(a-r) + (b-r) = c$. Solving this last equation for r gives:

$$r = \frac{a+b-c}{2} \quad (3)$$

A *primitive Pythagorean triple* (a, b, c) is a triple of positive integers in reduced form satisfying $a^2 + b^2 = c^2$; it gives the side lengths of a *primitive Pythagorean triangle*, where c is the hypotenuse and a, b are the legs. What can we say about the evenness or oddness of the sides when they have no common factor? Since $a^2 + b^2 = c^2$, taking parities mod 2 gives $a^2 + b^2 \equiv c^2 \pmod{2}$, so the combined parity of a, b must match the parity of c . This rules out three of the six naive combinations and leaves:³

²A triple of integers, (a, b, c) is in reduced form if they have no common integer divisor.

³We list the results of all combinations of even and odd numbers with respect to: addition, subtraction, and multiplication:

1. $E \pm E = E$; $E \times E = E$;

- c is even AND *exactly* one of the following is true:
 - Both a and b are even;
 - Both a and b are odd;
- c is odd AND exactly one of a and b is even (the other being odd).

Since the triple (a, b, c) is in reduced form, we can eliminate the case of a , b , and c all being even. Eliminating that case leaves the following two possibilities:

- c is even AND both a and b are odd;
- c is odd AND exactly one of a and b is even (the other being odd).

In both cases $a + b - c$ is even (odd + odd – even, or odd + even – odd), so equation (3) implies that the radius of the circle, r , is an integer. Using this fact and equation (2) we see that $\frac{ab}{2}$ must be an integer, which means the product ab is even. This knowledge eliminates the first choice; namely, that c is even with both a and b odd. Consequently, c is odd, with exactly one of a and b even (the other being odd).

The whole elimination tournament can now be seen at a glance. Writing E for even and O for odd, there are six possible parity patterns for (a, b, c) (up to a swap of the legs – we write the mixed-leg cases with b even); each losing pattern is eliminated by a specific weapon, and exactly one pattern survives:

a	b	c	fate
E	E	O	eliminated: parities must match in $a^2 + b^2 = c^2$
O	O	O	eliminated: parities must match in $a^2 + b^2 = c^2$
O	E	E	eliminated: parities must match in $a^2 + b^2 = c^2$
E	E	E	eliminated: the triple is in reduced form
O	O	E	eliminated: the area equation forces ab even
O	E	O	survives — the only possibility

2. $E \pm O = O$; $E \times O = E$;

3. $O \pm O = E$; $O \times O = O$.

These equations are to be read: An even integer plus (or minus) an even integer is an even integer. Likewise, an even integer plus (or minus) an odd integer is an odd integer, etc. These results are used throughout the rest of the document.

Following the surviving row, we assume (without loss of generality) that b is even; in which case, a is odd.

Now, the left hand side of (2) simplifies to $r(a + b - r) = r(r + c)$ by (3); with c odd, this is even regardless of r 's parity. This implies that $\frac{ab}{2}$ is even. Since b is even, we can write this fraction as: $a\frac{b}{2}$. So we know that $a\frac{b}{2}$ (the product of two integers) is even. And we know that a is odd; therefore, $\frac{b}{2}$ is even. This means that b is divisible by 4.

In summary, we have deduced the following properties for primitive Pythagorean triangles.

Proposition 1. *Let (a, b, c) be a primitive Pythagorean triple. Then:*

- *One of the legs is odd; the other leg is even; and the hypotenuse is odd.*
- *The even leg is divisible by 4.*
- *The radius, $r = (a + b - c)/2$, of the inscribed circle in the associated Pythagorean triangle is an integer (necessarily positive).*

Remark (what is “alternative” here). The standard routes to the first two facts are arithmetic. One goes through Euclid’s parametrization of primitive triples, $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$ with $m > n$ coprime and of opposite parity, from which divisibility of the even leg by 4 is immediate (in $2mn$, one of m and n is even). Another works modulo 8: odd squares are congruent to 1 mod 8, so with a and c odd, $b^2 = c^2 - a^2 \equiv 0 \pmod{8}$, which forces $4 \mid b$. The proof above replaces both with a single strategic move — *compute one quantity two ways*. The area of the triangle, reckoned whole and reckoned piece by piece, changes the representation of the algebraic identity $a^2 + b^2 = c^2$ into the integrality of the inradius $r = (a + b - c)/2$, and elementary even/odd bookkeeping does the rest. The dividend of the geometric route is the third conclusion — the integer inradius — which the arithmetic proofs never see, because their machinery never draws the circle.

The freshman project paid off after all: with no machinery beyond a dissected figure and parity arithmetic, the area accounting reproduced two classical theorems and added a third fact for free. And the story continues from here. Knowing now that the hypotenuse of a primitive triple must be odd, one can ask *which* odd numbers actually occur as hypotenuses, in how many ways, and how the prime hypotenuses are distributed among the primes. Those questions — one of which remains open — are the subject of the companion note, *A Conjecture on Primitive Pythagorean Triples* (`tex/conj_pythag_triple.tex` in this repository).

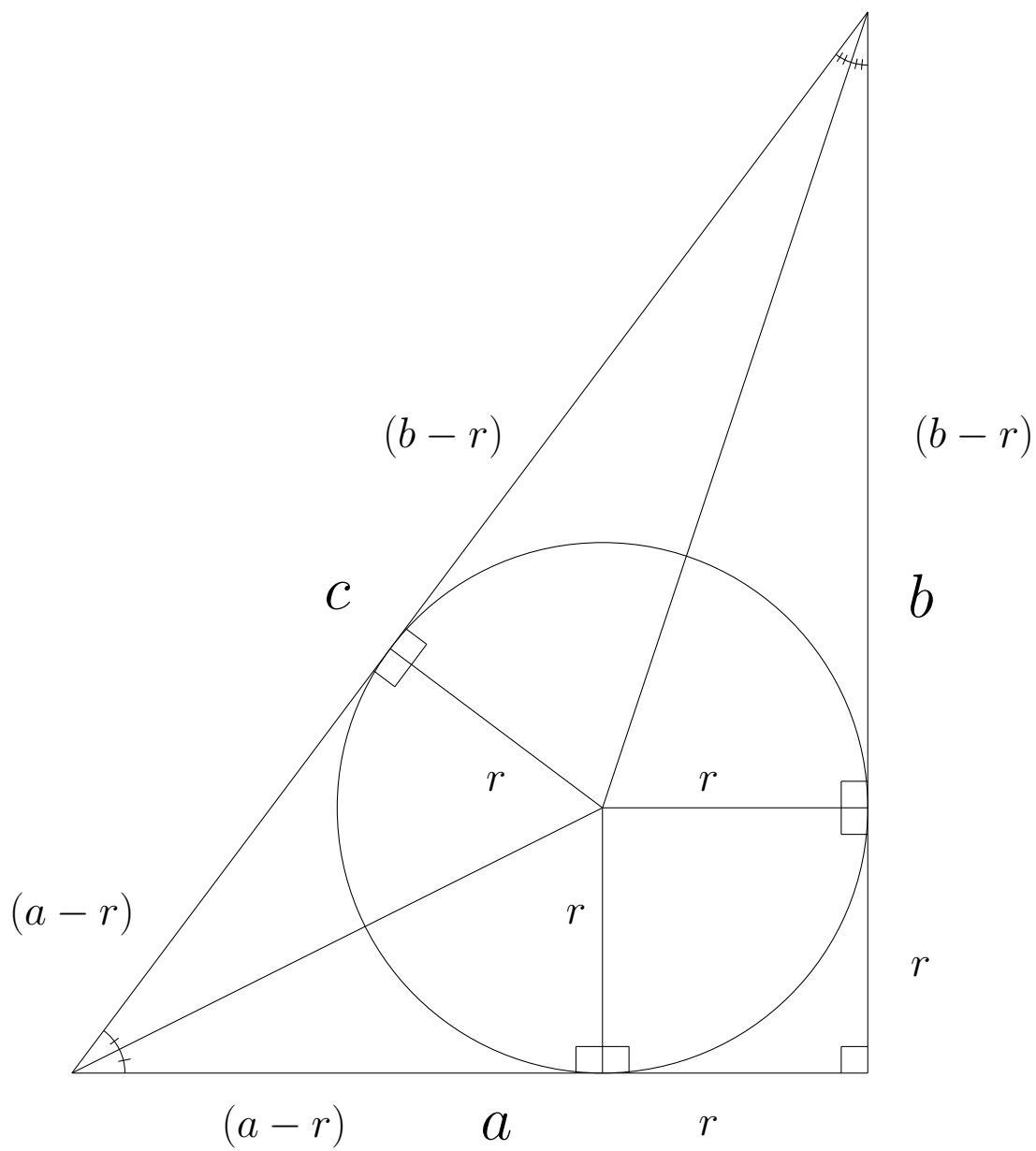


Figure 1: Dissected Right Triangle